	Target engagement	Dose	Efficacy	Safety
#15	**hepatitis C** HCV NS5A inhibitors (10)	**2011** HCV subgenomic replicon (Huh-7 cells; genotype 1a/1b and patient-derived NS5A hybrid…	**2015** HCV subgenomic replicon (Huh-7 cells; genotype 1a/1b and patient-derived NS5A hybrid…	**2020** HCV viral-kinetic mathematical model (Perelson/Dahari lineage) fitted to…	**2015** Repeat-dose toxicology in… Rat/monkey adrenal signal absent in patients
	hepatitis C HCV NS5B (non-nucleoside) inhibito…	**2014** HCV genotype 1 subgenomic replicon (Huh-7), with and without human serum	**2014** Rat and dog repeat-dose toxicology (dasabuvir)	**2018** Human-hepatocyte chimeric (uPA/SCID) mouse infected with… ■ Filibuvir add-on failed SVR desp…	**2014** Rat and dog repeat-dose toxicology (dasabuvir)
	hepatitis C HCV NS5B (nucleotide) inhibitors (7)	**2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2013** Rat, dog and mouse repeat-dose toxicology (GS-9851 / sofosbuvir IND package)	**2013** HCV subgenomic replicon (Huh-7 clone A; genotype 1b and chimeric 1-6 replicons)	**2016** Rat, dog and mouse… Tox missed BMS-986094 heart failure
#17	**acute myeloid leukaemia** FLT3 inhibitors (4)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2013** FLT3-ITD cell lines and MV4-11 xenografts (phospho-FLT3)	**2012** Ba/F3 FLT3-ITD resistance screen ■ Combination: no survival gain	**2018** Rat and dog toxicology and safety pharmacology PRES and QT missed by tox species
#19	**MASH** NR1H4 agonists (4)	**2019** Rodent FXR target-gene PD (tropifexor)	**2019** STAM and AMLN-diet NASH mice (tropifexor)	**2021** STAM and AMLN-diet NASH mice (tropifexor) □ Outcome trial negative	**2024** High-fat-diet mouse ALT/cholesterol screen ■ Cilofexor: no lipid change
	MASH FGF21 analogue (FGFR1c/KLB)…	**2023** Obese/diabetic cynomolgus monkeys	**2023** Pegozafermin population PK/PD exposure-response model… ■ Reached, failed fibrosis endpoin…	**2023** Pegozafermin population PK/PD exposure-response model… ■ Pegbelfermin failed fibrosis…	**2020** Diet-induced obese mice given recombinant human FGF21… ■ Mouse bone loss not seen at 16…
#20	**prostate cancer** AR antagonists (5)	**2013** LNCaP/AR castrate xenografts	**2013** LNCaP/AR castrate xenografts	**2016** LNCaP/AR castrate xenografts	**2019** Mouse brain:plasma distribution with rat and dog CNS toxicology (darolutamide)
#22	**ovarian cancer** PARP1/2 inhibitors (6)	**2009** Human tumour xenografts, needle-biopsy PAR ELISA (veliparib)	**2021** NOVA exposure–safety model (niraparib)	**2016** BRCA-deficient cancer cells and xenografts (niraparib)	**2017** Dog cardiovascular safety pharmacology (niraparib)
#23	**rheumatoid arthritis** BTK inhibitors (8)	**2013** BTK occupancy probe in CIA mice	**2022** CIA occupancy-efficacy PK/PD Target reached, no effect	**2019** Collagen-induced arthritis (rodent) CIA-active drug failed Phase 2	**2021** Human platelet aggregometry and humanized-mouse thrombus formation (acalabrutinib vs ibrutinib)
	rheumatoid arthritis SYK inhibitors (3)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457)	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) ■ Target reached, DAS28 unmove…	**2014** Rat adjuvant- and collagen-induced arthritis (MK-8457) ■ DAS28 no better than placebo	**2018** Cynomolgus monkey platelet/bleeding-time (lanraplenib) ■ Serious infections missed,…
#24	**multiple myeloma** TNFRSF17-directed depleters (8)	**2016** BCMA+ myeloma xenografts in NSG mice, CAR-T	**2014** BCMA+ myeloma xenografts in mice, ADC	**2019** bb2121 CAR T cytotoxicity against cell lines of quantified BCMA density (in vitro)	**2020** Rat and rabbit repeat-dose toxicology (belantamab mafodotin)
#27	**open-angle glaucoma** ROCK inhibitors (3)	**2018** Living mouse eye, OCT of conventional outflow tissue, topical netarsudil	**2015** Normotensive monkey eye, topical netarsudil 0.04%	**2015** Normotensive monkey eye, topical netarsudil 0.04% ■ Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
All.	**allergic rhinitis** allergen (immunotherapy) (8)	**2018** Conjunctival provocation test (CPT), gp-ASIT+	**2019** Conjunctival provocation te… CPT dose reached, confirmatory trial missed	**2019** Conjunctival provocation te… Confirmatory field trial missed primary	**2019** Allergen exposure chamber (EEC)
Autoimmune	**psoriasis** RORC inverse agonists (4)	**2021** Human whole-blood IL-17 secretion assay ■ No IL-17 change in plaques	**2021** Human whole-blood IL-17 secretion assay	**2022** IL-23 minicircle mouse, oral BI 730357	**2015** Nonclinical toxicology species (not public)
	psoriasis anti-IL17A antibodies (4)	**2016** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** Cynomolgus repeat-dose toxicology	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout	**2016** People with inborn errors of IL-17 immunity
	ulcerative colitis S1PR1 agonists (3)	**2016** Rodent blood lymphocyte counts (ozanimod)	**2023** Rat and dog repeat-dose toxicology (etrasimod)	**2016** TNBS and CD4+CD45RBhi transfer colitis mice (ozanimod)	**2023** hERG current assay in vitro (etrasimod) ■ Ozanimod still slows the heart
GI	**GERD** ATP4A blockers (3)	**2019** Stimulated gastric acid secretion in dogs	**2017** Stimulated gastric acid secretion in dogs	**2019** Rat reflux oesophagitis vs esomeprazole	**2025** Rat repeat dosing: gastrin, emptying, microbiota
Hematology	**hemophilia A** F8 replacement / gene therapy (8)	**2025** Cynomolgus macaques with pre-existing anti-AAV5 antibodies given valoctocogene roxaparvovec…	**2017** F8-knockout mice + cynomolgus macaques	**2021** Hemophilia A dogs (Chapel Hill / Queen's colonies) ■ Mouse-stable BAX 888 lost in all…	**2023** Hemophilia A dogs (Chapel Hill / Queen's colonies)
	hemophilia B F9 replacement / gene therapy (7)	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua ■ FIX expression lost in 7 of 8	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua	**2024** Juvenile hemophilia B dogs, canine surrogate vector	**2024** Cynomolgus GLP tox, fidanacogene predecessor
	PNH anti-C5 antibodies (4)	**2022** Hemolysis assay with C5 R885H variant	**2020** Cynomolgus PK/PD (crovalimab)	**2018** Complement hemolysis assay in human serum	**2020** Drug–target–drug complex assay, human C5
Met.	**hyperlipidemia** CETP inhibitors (3)	**2014** Human CETP/apoA-I transgenic mouse	**2014** CETP/apoA-I mouse; rat blood pressure ■ Exposure far below prediction	**2011** Human CETP loss-of-function genotypes □ Outcome trial negative	**2014** Rat blood pressure + human adrenal H295R cells
Psy.	**insomnia** HCRTR1/2 antagonists (6)	**2013** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2022** PBPK-PD model of brain OX2 receptor occupancy (daridorexant, 65% occupancy threshold)	**2012** Rat, dog and rhesus sleep EEG/EMG (suvorexant)	**2019** Mouse emotion-induced cataplexy (lemborexant) ■ Animals no abuse; humans liked it
Respiratory	**cystic fibrosis** CFTR correctors (10)	**2024** FRT cells expressing CFTR variants ■ Third corrector added nothing	**2015** Rat and dog toxicology (lumacaftor)	**2024** FRT cells expressing CFTR variants	**2025** Rat and dog tox (vanzacaftor)
	cystic fibrosis CFTR potentiators (5)	**2026** Patient-derived primary nasal epithelial cells (HNE, Ussing chamber theratyping)	**2024** Patient-derived HBE cells (Ussing chamber)	**2024** FRT cells carrying each CFTR variant (Ussing chamber)	**2024** Rat toxicology (juvenile and adult, ivacaftor program)
Uro.	**interstitial cystitis** anti-NGF antibodies (3)	**1997** Rodent chemical cystitis	**1997** Rodent chemical cystitis	**2022** ASP6294 vs anti-NGF IgG in arthritic rodents ASP6294 missed IC pain endpoint	**2012** Primate and rodent joint tox Joint harm missed; IC trials put on hold
Other	**atopic eczema** PDE4 inhibitors (5)	**2019** Stimulated human blood cells (PBMC cytokine release)	**2016** Minipig dermal toxicity (3- and 9-month)	**2023** Mouse chronic allergic contact dermatitis (difamilast) ■ Explant engaged; primary failed	**2016** Rabbit primary skin irritation